

JESS Thermodynamic Database v8.9

Francium

24-Jan-23

Reaction No. 71105							
$\text{Fr}+1 + e-1 = \text{Fr}(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:70.500 (5SF)	0	CRV	9029
Note (1): Low wgt for internal consistency							
2	25	0	Inf. Dilution	E:-2.9 (2SF)	4	CRV	6330
3	25	0	Inf. Dilution	E:-2.9 (2SF)	4	CRV	7761
4	25	0	Inf. Dilution	dH:245.3 (4SF) kJ	3	CRV	7761
5	25	0	Inf. Dilution	dH:62.500 (5SF)	0	CRV	9029

Reaction No. 71167							
$\text{Fr}+1 + 0.5\text{H}_2(g) + e-1 = \text{FrH}(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.75 (3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:204.9 (4SF) kJ	3	CRV	7761

Reaction No. 71168							
$\text{Fr}+1_{\text{OH}-1} + e-1 = \text{Fr}(s) + \text{OH}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.97 (3SF)	3	CRV	7761

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt

3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review

References

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K

Reference No.: 9029(97SSW)

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